Step 1 (v3): Final Parameters

After Integrating Ji et al. (2019), *Nucl. Eng. Des.* 355

Particle Bed Reactor Stability Simulator

SwampHacks XII Project

What this version does. The uploaded optimization paper [J19b] was read in full. It supplies the missing inlet conditions, an explicit outlet pressure, three published axial power shapes, design limits, the full setup of the earlier optimization, and new quantitative validation targets. Every affected decision was re-checked for quality and hardcoding risk using the same process as v2. Marks: ✓ unchanged, ✓ mod modified, ✓ up upgraded from assumption to sourced. This version **replaces** v2 as the freeze document.

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1 What the New Paper Changed

#	Decision	Status	Change
D1	Inlet temperature and pressure	✓ up	Inlet 200 K and chamber (outlet) pressure 7.0 MPa are now stated directly for a fuel element [J19b].
D2	Mass flow by energy balance	✓	New evidence shows the energy mismatch is systematic in the Tsinghua property model, not a misreading. Rule kept.
D4	Correlations	✓	[J19b] uses the same three correlations; ranges recomputed for the new nominal element.
D7	Fuel heat capacity	✓ mod	ZrC density 6.78 g/cm^3 added from a supplier data sheet [HM]; particle composition from [J19b] noted.
D9	Study ranges and optimizer	✓ mod	Design limits now sourced (3800 K, 600 kPa); optimizer runs a reproduction mode using the paper's exact setup plus our mode.
D10	Validation criteria	✓ mod	Four new targets from [J19b] (V10–V13).
D11	**Nominal element** (new)	✓	**Switched to the [J19b] preliminary design**; the [T19] element moves to validation.
D12	**Axial power shapes** (new)	✓	Three published shapes adopted; normalization verified.

D3, D5, D6, and D8 are unaffected and remain checked off as in v2.

> **The most important finding for your pitch.** In its conclusions, [J19b] lists as future work: replacing CFD with an accurate surrogate model (CFD limited their optimizer to small populations), adding nuclear feedback, and quantifying more uncertain parameters. It also optimizes at a single operating condition. Your project directly answers the first item and adds something new: **designs must stay stable across the 20%–100% throttle range**, which the earlier optimization does not consider. Phrase it that way rather than claiming to be first.

2 Key Data From Ji et al. (2019)

2.1 Design requirements (their Table 1)

Parameter	Value
Inlet temperature	200 K
Expected outlet temperature	3000 K
Expected chamber pressure	7.0 MPa
Mean power density	20.0 MW/L
Mass flow rate	0.505 kg/s
Temperature limit	3800 K
Pressure-drop limit	600 kPa

The 200 K inlet is described as a common value after preheating by the regenerative nozzle and radial reflector; 3800 K is the melting point of the ternary carbide fuel (U, Zr, Nb)C; 600 kPa is about one third of the NERVA core pressure drop [J19b].

2.2 Preliminary fuel element geometry (their Table 2)

Parameter	Value
Hot gas channel radius	25 mm
Outer radius of hot frit	27 mm
Outer radius of particle bed	37 mm
Outer radius of cold frit	39 mm
Fuel element radius	41 mm
Length	0.65 m
Particle diameter; bed porosity	500 μm; 0.38
Cold frit pore size; porosity	2 μm; 0.5
Hot frit pore size; porosity	100 μm; 0.5

Derived: frits 2 mm each, bed 10 mm (20 particle diameters), plenum 2 mm; bed volume 1.307 L; power at 20 MW/L = 26.14 MW. The paper uses the Z flow pattern.

2.3 Published results (quantitative targets)

Case (preliminary design, 0.505 kg/s)	$T_{\max}$ **(K)**	Δp **(kPa)**
Uniform power	3348.4	529.8
Symmetric cosine power	3905.5	–
Asymmetric power	3884.1	530.6
Power with top and bottom reflectors	3830.8	529.4
Optimized (Pareto point chosen)		
A: cosine, $\{b, c, \alpha_0\} = \{0.434, 1.070, 0.533\}$	3147.6	414.4
B: asymmetric, $\{0.444, 1.052, 0.582\}$	3151.8	312.0
C: reflectors, $\{0.418, 1.036, 0.478, 0.164, 28.470, 0.162, 32.830\}$	3287.9	562.4

For point A, the porosity peak sits upstream of the power peak, with lower porosity near the outlet, which the authors attribute to the hot channel pressure drop and flow shift [J19b].

2.4 Optimizer setup used by the paper

- Algorithm: NSGA-II; objectives minimize peak temperature and pressure drop, subject to the 3800 K and 600 kPa limits.
- Design variable: the cold frit porosity profile, $\varepsilon = \varepsilon_0 + \varepsilon_1 + \varepsilon_2$, a parabola times mean porosity α_0 plus two exponential terms near the top and bottom reflectors (7 parameters $b, c, \alpha_0, \alpha_1, \lambda_1, \alpha_2, \lambda_2$), with the mean relative porosity fixed at 1.
- Bounds (bare element): $b \in [0.3, 0.5]$, $c \in [1.0, 1.2]$, $\alpha_0 \in [0.4, 0.6]$, $\alpha_1 = \alpha_2 = 0$. For the asymmetric case $b \in [0.4, 0.7]$. With reflectors: $b \in [0.4, 0.5]$, $\alpha_{1,2} \in [0.1, 0.4]$, $\lambda_{1,2} \in [10, 50]$.
- Population 8 (16 with reflectors); crossover probability 0.9; mutation probability 0.5; converged in about 10–12 generations.
- Each evaluation is a CFD run (ANSYS FLUENT), which limited the population size.

2.5 Uncertainty analysis used by the paper

Normal distributions with two-standard-deviation ranges: inlet temperature ±20 K, mass flow ±5%, power density ±2.5%, cold frit pore size ±5%; 220 Latin hypercube samples for two-sided (95%, 95%) tolerance limits. Results for optimized point A: peak temperature 2986–3369 K, pressure drop 368.4–464.3 kPa, outlet temperature 2841–3171 K [J19b]. This is a ready-made Tier 3 benchmark.

3 Decision Updates

3.1 D1: Inlet temperature and pressure ✓ up

Before: 200 K and 8.0 MPa borrowed from another case, with ambiguity about where the pressure applies.

Now: for the new nominal element (D11), [J19b] states inlet temperature 200 K and expected *chamber* pressure 7.0 MPa. Chamber pressure is, by definition, the pressure at the element outlet, so the location question is resolved for the nominal case.

Hardcoding check: both values are sourced and fixed before runs. The outlet/inlet sensitivity pair from v2 now applies only to the [T19] validation cases, where the ambiguity remains.

3.2 D2: Mass flow by energy balance ✓

New evidence: with JANAF data, [J19b]'s inputs (26.14 MW, 0.505 kg/s, 51.8 MJ/kg) imply an outlet near 3340 K, while the paper expects 3000 K. The [T19] element showed the same pattern (51.2 MJ/kg for about 3000 K). In [T19]'s larger reactor, 47.6 MJ/kg gave 2772 K. All three are consistent with an enthalpy rise roughly 13–15% higher than NIST-JANAF in the Tsinghua hydrogen model. The discrepancy is therefore most likely systematic in their property model, not an error in how we read the papers.

Is the rule still best? Yes. Matching their property model to agree with them would be tuning toward the target. NIST-JANAF is the authoritative ideal-gas reference [JANAF].

Consequences to state up front:

- The rule gives about 0.576 kg/s for the nominal element (hand estimate), about 14% above the published 0.505 kg/s.
- In validation runs using published inputs, our temperatures will run several hundred kelvin hotter than theirs. Temperature-based criteria therefore compare *directions and changes*, not absolute values (Section 6).

3.3 D4: Correlations ✓

[J19b] uses Ergun for the frits (pore size and porosity), KTA 3102 for bed pressure drop, and KTA 3102 for heat transfer, the same as our choice. Reynolds numbers for the new nominal element (NASA viscosities):

Power	Re, cold face	Re_m, cold	Re, hot face	Re_m, hot
100% (0.576 kg/s)	279	451	57	92
20%	56	90	11.5	18.5
5%	14	23	2.9	4.6

Same conclusions as v2: pressure drop in range except at the hot face below about 15% power; heat transfer below its Reynolds range on the hot side. The bed is exactly 20 particles thick, at

the edge of the heat transfer correlation's geometric limit. Hot channel Reynolds about 3.2×10^5 (100%) and 6.4×10^4 (20%); plenum about 6.7×10^5; all within the friction correlation's range.

3.4 D7: Fuel heat capacity

✓ mod

Added: ZrC density 6.78 g/cm^3, from the Heeger Materials ZrC powder data sheet [HM]. (An earlier note said 6.73; the cited page lists 6.78, so 6.78 is used. The 0.7% difference is far inside the ±50% sensitivity range.) This is a supplier page, not peer-reviewed; the ZrC review [KA13] can replace it later. Volumetric heat capacity of ZrC: about 3.5, 3.8, and 4.0 MJ/(m^3·K) at 1000, 2000, and 3000 K; about 2.2–2.5 MJ/(m^3·K) averaged over the bed (solid fraction 0.62).

Composition note: [J19b]'s Figure 2 shows each particle as a UC_x kernel, porous graphite, high-density graphite, and a ZrC coating. A pure-ZrC proxy is therefore uncertain; the ±50% sensitivity case covers it. As shown in v2, this affects only oscillatory instabilities and time scales, not the static instability boundary.

3.5 D9: Study ranges and optimizer

✓ mod

Changes:

1. **Sourced design limits:** peak temperature ≤ 3800 K and pressure drop ≤ 600 kPa [J19b] become the optimizer constraints (replacing our unsourced ones). App warning thresholds: 2800 K (SNTP baseline fuel), 3100 K, 3500 K [SNTP], and 3800 K [J19b].
2. **Design variable changed to porosity,** matching [J19b]'s physics, instead of a generic resistance multiplier.
3. **Two optimizer modes:**
   - *Reproduction mode:* the paper's 7-parameter shape function, bounds, objectives, and limits exactly, so results are directly comparable to Pareto points A, B, and C.
   - *Our mode:* 8 axial zones with free porosity per zone, plus the stability constraint (stable at every power from 20% to 100%, mass flow scaled with power).
4. **Free-zone bounds by rule, not by hand:** each zone's porosity is limited to the range the paper's shape function can reach within its published bounds (computed in code).
5. **Fair comparison:** our optimizer gets the same evaluation budget as the paper's run (about 100–200 evaluations); we report results at that budget and at a larger one.

Power sweep: 5%–100% of nominal is now 1–20 MW/L, still bracketing the NASA threshold of about 4 MW/L [K93]; 20% power is exactly 4 MW/L.

Hardcoding check: every bound, limit, and objective now comes from a source or a stated rule.

3.6 D10: Validation criteria

✓ mod

Four targets added (V10–V13); full list in Section 6.

3.7 D11 (new): Nominal fuel element ✓

Decision: use the [J19b] preliminary design as the nominal element; keep the [T19] element for validation targets V5 and V6.

Why it is best: [J19b] states every input we previously had to assume (inlet temperature, outlet pressure, power density, flow, limits, frit properties), publishes results for four power shapes, documents a complete optimization to compare against, and provides uncertainty results. It is the best-documented element available.

Hardcoding check: the switch happens before any code or run, based on documentation quality, not on results.

3.8 D12 (new): Axial power shapes ✓

Shape	Formula [J19b]	Our check
Uniform	$F = 1$	–
Symmetric cosine	$F = 1.25\cos(\pi(x - L/2)/(1.4L))$	Mean 1.004; peak 1.25
Asymmetric	$F = 0.398 + 1.9752\,s + 0.5265\,s^2 - 2.244\,s^3$, with $s = x/L$	Mean 1.000; peak 1.29 at $s = 0.63$
With reflectors	$F = \sum_{i=0}^{6} a_i s^i$ with a_0 to $a_6 = 1.094$, -4.002, 31.445, -101.605, 166.788, -138.205, 45.517	Mean 0.999; range 0.90–1.09

All three non-uniform shapes average to 1 within 0.4%, confirming they were transcribed correctly. **Rule:** renormalize each shape to a mean of exactly 1 in code so total power is unchanged. **Tier 1:** uniform and cosine; **Tier 2:** asymmetric and reflector shapes.

4 Anti-Hardcoding Protocol (Unchanged, Plus Two Rules)

The eight rules from v2 still apply. Two are added:

9. **Reproduction mode uses the paper's settings exactly**: shape function, bounds, objectives, limits, population, crossover, and mutation probabilities.
10. **Equal evaluation budgets** when comparing our optimizer with the paper's.

5 Final Freeze Table

Status key: **S** sourced; **A** assumption; **R** rule computed in code; **D** design choice.

5.1 Nominal fuel element ([J19b] preliminary design)

Parameter	Value or rule	Status	Note
Hot channel radius	0.025 m	S	[J19b] Table 2
Hot frit outer radius	0.027 m	S	
Bed outer radius	0.037 m	S	
Cold frit outer radius	0.039 m	S	
Element (plenum outer) radius	0.041 m	S	
Length	0.65 m	S	
Particle diameter; bed porosity	0.0005 m; 0.38	S	
Cold frit pore size; porosity	2 μm; 0.5	S	
Hot frit pore size; porosity	100 μm; 0.5	S	
Power density (bed volume)	20.0 MW/L	S	26.14 MW
Inlet temperature	200 K	S	[J19b]
Outlet (chamber) pressure	7.0 MPa	S	[J19b]
Target outlet temperature	3000 K	S	
Mass flow	Energy-balance rule (D2)	R	$\approx$0.576 kg/s estimate
Mass flow, published	0.505 kg/s	S	Validation runs; sensitivity
Throttling	Mass flow $\propto$ power	D	
Flow pattern	Z (baseline); U	S	
Axial power shapes	D12 table	S	Renormalized to mean 1
Design limits	3800 K; 600 kPa	S	[J19b]

5.2 Validation elements ([T19])

T19 fuel element (V5, V6): length 0.6392 m; hot channel radius 21.29 mm; hot frit 3.00 mm; bed 6.15 mm; cold frit 2.00 mm; plenum 3.00 mm; particles 0.5 mm; porosity 0.38; cold frit 2 μm / 0.4; hot frit 0.1 mm / 0.5; 30 MW/L; 0.396 kg/s; inlet 200 K and pressure 8.0 MPa (assumptions, run outlet- and inlet-referenced). Hot-frit-dominated variant by the D8 rule.

T19 larger reactor (V1–V4): as listed in v2: 507 mm radius; 1200/1000/800 mm lengths; hot channel 100 mm; frits 3 and 2 mm; bed 281 mm; plenum 10 mm; 6 mm particles; 4 mm frit pores; porosity 0.38, 0.1, 0.2; 1000 MW; 200 K; 21/17.5/14 kg/s; 8.0 MPa.

5.3 Properties, correlations, fuel

Unchanged from v2 (normal hydrogen; CoolProp below 1000 K; NIST-JANAF and NASA transport coefficients above; smooth blend 950–1050 K; KTA and Ergun with Wakao–Kaguei and Ergun sensitivity cases; Petukhov friction; manifold momentum). Fuel: ZrC proxy, c_p from NIST-JANAF, density 6.78 g/cm^3 [HM], ±50% sensitivity.

5.4 Study settings

Item	Value
Power sweep	5%–100% (1–20 MW/L); 20%–100% labeled operational
Frit share sweep	Continuous, near zero to design value
Optimizer, reproduction mode	[J19b] shape function, bounds, NSGA-II settings, limits
Optimizer, our mode	8 zones; porosity bounds by rule; Pareto (peak temperature, pressure drop); limits 3800 K and 600 kPa; stable over 20%–100%
Evaluation budget	Matched to [J19b] (∼100–200), plus a larger run
Uncertainty analysis (Tier 3)	[J19b] inputs and distributions; tolerance limits with 220 samples

6 Validation Criteria (Pre-Registered)

All runs use the source's stated inputs. Because of the systematic enthalpy difference (D2), temperature criteria compare directions and relative changes.

ID	Target	Pass criterion
V1	T19 larger reactor, Z: Δp = 872.4 kPa	Within ±25%
V2	Z vs. U: peak temperature falls; Δp nearly unchanged	U lower peak temperature; Δp changes by less than 2%
V3	Core heights 80/100/120 cm	Δp within ±25% each; β decreases with shorter core
V4	Frit-dominance variants	Both lower peak temperature vs. baseline
V5	T19 element, cold-frit-dominated: Δp = 1189.2 kPa	Within ±25%; stable
V6	T19 element, hot-frit-dominated (D8 rule)	Instability, or no steady solution below 8000 K
V7	No cold frit at design exit temperature [K93]	Static instability below a power density within a factor of 3 of 4 MW/L
V8	Strong cold frit [K93]	V7 instability disappears

ID	Target	Pass criterion
V9	Ledinegg check	Eigenvalue and pressure-drop-slope onsets agree within one sweep step
V10	J19b preliminary design, uniform power, 0.505 kg/s: $\Delta p = 529.8$ kPa	Within ±25%
V11	Cosine vs. uniform power: peak temperature rises (3348.4 to 3905.5 K); Δp essentially unchanged	Peak temperature rises; Δp changes by less than 2%
V12	J19b optimal point A applied, cosine power: peak temperature and Δp both fall (to 3147.6 K and 414.4 kPa)	Both fall; Δp within ±25% of 414.4 kPa
V13	Reproduction-mode optimizer, cosine power	Chosen design has porosity peak upstream of the power peak ($b < 0.5$), as in [J19b]

Expected disagreements to report, not fix: absolute temperatures (D2); 1D model vs. 2D CFD; parallel-passage limitation at low Reynolds number [K93]; [J19b]'s CFD treats the flow as incompressible, while our model includes density change.

7 What Code Must Settle (Rules Fixed)

Item	Rule
Exact nominal mass flow	D2 formula with the property module
Hot-frit-dominated frits (T19 element)	D8 swap-shares rule
Free-zone porosity bounds	Range reachable by the [J19b] shape function within its bounds
Power shape normalization	Renormalize to mean exactly 1
Grid size	Start 20 slices and 20 radial steps; double until peak temperature and Δp change by under 1%
Property continuity	Values and slopes continuous over 900–1100 K
Ideal-gas error at 1000 K	Compare with CoolProp; report
Correlation range flags	Per run

8 Remaining Items and Readiness

8.1 Papers you could not find

- **Ji et al. 2020 (*Nuclear Technology*):** was needed for inlet conditions and frit studies. [J19b] now supplies the inlet conditions, so this paper is **no longer blocking**. If you want it later, request it through UF Libraries interlibrary loan.
- **Ludewig et al. 1996:** was needed for Brookhaven design details and power shapes. [J19b]'s geometry and power shapes cover this, so it is also **no longer blocking**.

8.2 Checklist before coding

- ✓ ZrC density source recorded [HM].
- ☐ Copy the NASA hydrogen coefficients directly from the report PDF and double-check them.
- ☐ Download the NIST-JANAF H_2 and H tables as data files.
- ☐ Confirm the Wakao–Kaguei validity range in its source.
- ☐ Have a UF Nuclear Engineering faculty member review D2 (systematic enthalpy difference), D8, D11, and the validation criteria.

> **Ready to code.** After the checklist, every parameter, rule, criterion, and protocol is fixed. The first code files are `params.py` (this document's freeze table), `properties.py` (D3), and the verification tests. Commit them before the first validation run.

Sources

[J19b] Y. Ji, H. Zhang, J. Sun, L. Shi. Thermal performance optimization of a fuel element in particle bed reactors for nuclear thermal propulsion. *Nucl. Eng. Des.* 355:110316, 2019. doi:10.1016/j.nucengdes.2019.110316

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[SV95] R. A. Svehla. Transport Coefficients for the NASA Lewis Chemical Equilibrium Program. NASA TM-4647, 1995. https://ntrs.nasa.gov/citations/19950021761

[KA13] Y. Katoh, G. Vasudevamurthy, T. Nozawa, L. L. Snead. Properties of zirconium carbide for nuclear fuel applications. *J. Nucl. Mater.*, 2013.

[HM] Heeger Materials. Zirconium Carbide Powder, ZrC, CAS 12070-14-3 (product data sheet; density 6.78 g/cm^3). https://heegermaterials.com/carbide-powder/1505-zirconium-carbide-powder.html

[v2] All other sources as listed in Step 1 (v2).